

SEAMLESS POWER

UNINTERRUPTED LIVING

LIGEN-INV2000 - 24 VDC

Solar Features

- Auto switchover between Solar and Main charging
- Zero drop solar charger (Max. 50A)
- Simultaneous charging through mains as well as solar panel
- Short charging time
- Long battery backup



EFFICIENT POWER PERFORMANCE

High efficiency with low energy consumption.

CRGO LAMINATED CORE

Premium quality for improved performance and durability.

PURE COPPER TRANSFORMER

Ensures optimal power conversion.

INVERTER
24V -2.0KVA

MPPT 50AMP/100Voc
1.25KW PV PANEL

GRID CHARGER
Max. 15AMP

- Zero Grid Power Consumption on Idle Stage
- Battery Grid Charging even at low grid Voltage
- Special charging algorithm to enhance battery life
- Intelligence charging adapts itself to low, high battery, more and less power cut areas, battery health condition.
- Forced Grid to Solar Mode Shifting Facility
- Constant Noise Free DSP Pure Sine Wave Output
- 4 Line Multifunctional LCD Front Panel display

- Intelligent true multi stage smart solar and grid charging (Bulk, Absorption, Float)
- High over load and surge load handling capacity
- MCB Short Circuit Protection for Grid Power
- Easy to operate
- No maintenance cost
- Fumeless



www.ligenpower.com



SPECIFICATIONS

Product Overview

The “LIGEN POWER” inverter is a robust commercial-grade solution designed for small to medium-sized businesses, offices, and shops. With its advanced features and sturdy build, it ensures uninterrupted power supply during outages, enabling smooth operations. Key highlights include:

Description	Details	Specification	Details
Model No	LIGEN-INV2000 – 24 VDC	Model	2.0KVA MPPT PCU Inverter
Product	Solar MPPT PCU Inverter	Rated Power	2.0KVA/1.6KW (1-Phase Input 1-Phase Output)
Functionality	Works with Grid, Solar & Integrated LFP Battery	Input Voltage (Battery Mode)	25.6V DC
DSP Based	Less components, compact size, reduced electricity bill, higher efficiency	BATTERY CHEMISTRY	Lithium iron phosphate (LFP)
Soft Start Features	Protects appliances during start-up	BATTERY PACK	INTERNAL INTEGRATED WITH 851P BMS
Usage	Reliable power backup for residential & commercial applications	BATTERY CAPACITY	100Ah
Technology	Built-in MPPT solar charge controller for optimized performance. Utilizes Sine Wave Technology for safe and efficient power delivery.	Grid Input Voltage (Vmp)	90V - 260V AC
Battery Support	Internal 25.6V LiB Battery with 851P BMS	Solar Input Voltage (Vmp)	36V - 75V DC
Intelligent Processing	DSP technology enables advanced control algorithms for precise output regulation, adaptive learning for optimized battery management, and fault detection for enhanced reliability. Efficient power management and quick fault response ensure uninterrupted operation in offline applications.	Solar Voc	≤ 100V DC
Smart Display	Offers unique indications for mains status, Output status, system faults, Load status, Temperature status, Charging and battery status.	Maximum PV Connected	1.25KW@8N
Operation Modes	Involves switching from mains power to battery power and Solar power during outages, ensuring uninterrupted supply to connected devices. Ideal for sensitive appliances like computers.	Solar Controller Type	MPPT (Maximum Power Point Tracking)
Protection Features	Includes MCB/Fuse for input mains protection and a bypass switch for direct main supply in case of inverter faults.	Charging Modes	Solar, Grid, Solar + Grid (If Solar CHG ≤ [5.0A±0.5A])
Grid Protection	AC input low and high voltage cut off protections in both, inverter and UPS modes	Output Waveform	Pure Sine Wave
Noiseless Operation	Low harmonic distortion ensures quiet operation. This inverter prioritizes reliability, efficiency, and safety, making it an ideal choice for commercial establishments requiring dependable backup power solutions.	Efficiency	> 90%
		Max Solar Charging Current	50A±5A (Boost Mosfet based)
		Max Grid Charging Current	15A±1A (CCCV)
		Operation Logic for Charging	Solar Priority (By default) / Grid Priority
		Output Voltage (Backup Mode)	220V AC ± 2%
		Output Voltage (Grid Mode)	Same as Input
		Output Frequency on Battery	50Hz ± 0.05Hz
		Changover Time	<10ms
		Protection	Yes (Automatic Cut-off)
		Protection Supported	Overload, Deep Discharge, Over Charge, Short Circuit, Over Temperature
		Over Load Capacity	>120%: 3 times auto reset, 4th time shut down / >200% Output goes down
		DISPLAY ON MPPT	ARRAY VOLTAGE, ARRAY CURRENT, BATT VOLTAGE, BATT CURRENT, CHG STATUS, KWH
		DISPLAY ON UPS/INVERTER	LOW BATT., BATT. VOLTAGE, CHG CURRENT, CHG MODE, AC IN, AC OUT, LOAD, OVERLOAD, TEMPERATURE, INV ON/OFF STATUS
		Operating Temperature Range	-20°C to 65°C
		Cooling Mechanism	Intelligent Fan Cooling
		Cooling Fan Start/Stop	60°C or Load above 50% / 50°C & Load below 50%
		Audible Alarm	Yes
		Bypass Feature	Manual